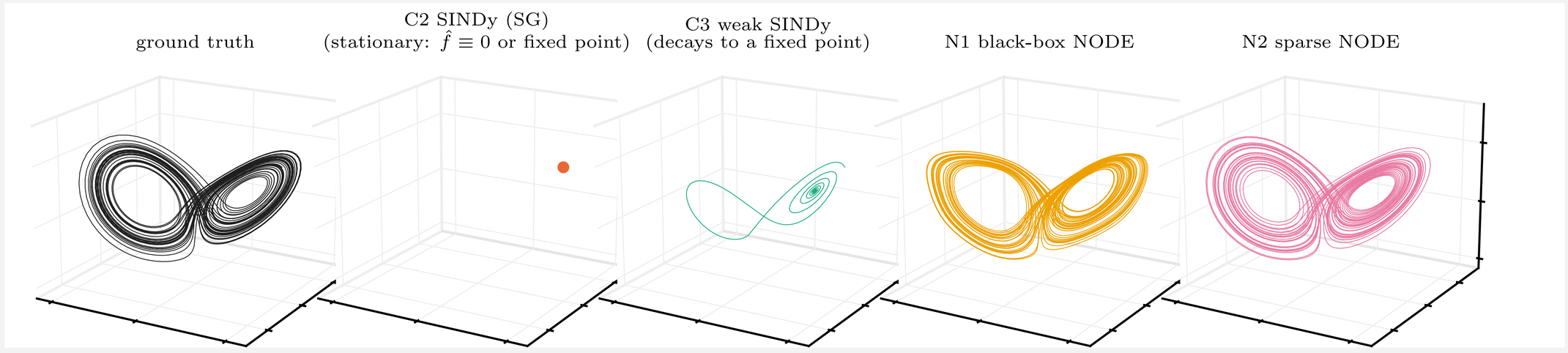
Learning and Discovering Chaotic Dynamics from Sparse, Noisy Observations

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Reconstructed attractors at coarse sampling: free runs of the identified models



Lorenz-63 sampled every $\Delta t_{\mathrm{obs}} = 0.2$ (about 4 samples per oscillation), $\varepsilon = 1\%$. Free runs over $60\, t_{\mathrm{char}}$. SINDy (SG) selects $f \equiv 0$, and weak SINDy decays to a fixed point. The black-box NODE stays near the attractor but its equations are wrong. The sparse NODE recovers

$$\dot{x} = -9.59x + 9.73y, \quad \dot{y} = 28.03x - 1.02y - 1.00xz, \quad \dot{z} = -2.66z + 1.00xy$$

(plus two terms of size ~ 0.02). Median coefficient error over seeds: **0.05**, against ≥ 2.4 for every classical method.

Question

Given a few short, noisy, coarsely sampled records of a chaotic system $\dot{x} = f(x)$, can we **recover the equations**, **forecast** up to the predictability limit, and **know when not to trust** the model? And when do neural ODEs beat sparse regression?

Problem formulation

Observations

$$y_k^{(e)} = x^{(e)}(t_k) + \eta_k^{(e)}, \qquad \eta \sim \mathcal{N}\big(0, \varepsilon^2 \Sigma_x\big),$$

on a grid of step Δt_{obs}, for $E = 4$ short experiments.

Sparse library model

$$f(x) = \boldsymbol{\Xi}^{\top} \theta(x), \quad \theta(x) = \big(1,\, x_i,\, x_i x_j\big)_{i \leq j}, \quad \boldsymbol{\Xi} \text{ sparse.}$$

Coordinates are scaled without centring, $z = S^{-1}x$, $\tau = t/t_{\mathrm{char}}$, so that $\tilde{\boldsymbol{\Xi}} = t_{\mathrm{char}}\, C \boldsymbol{\Xi} S^{-1}$ keeps the support.

Why trajectory matching is hard on chaos

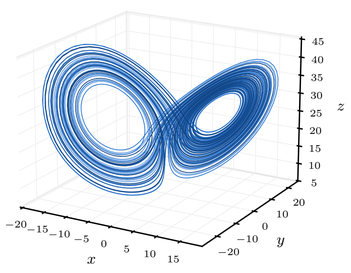
The sensitivity $U = \partial x / \partial \vartheta$ obeys $\dot{U} = J(x)U + \partial_\vartheta f$, so $\|U(t)\| \sim e^{\lambda_{\max} t}$. The single-shooting loss $\mathcal{L}_T$ gets rough, and its convex-like basin shrinks like $e^{-\lambda_{\max} T}$.

Multiple shooting with latent states

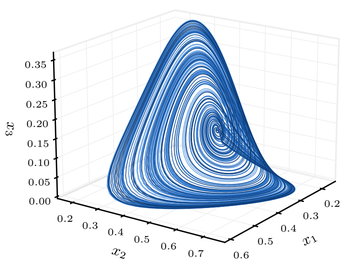
$$\mathcal{J}_K(\vartheta, S) = \frac{1}{|\mathcal{W}|K} \sum_{i,k} \big\| \Phi_\vartheta^{t_{i+k} - t_i}(S_i) - y_{i+k} \big\|_{\Sigma^{-1}}^2 + \frac{w}{N} \sum_i \|S_i - y_i\|_{\Sigma^{-1}}^2$$

Sensitivities are bounded by $e^{\lambda_{\max} K \Delta t_{\mathrm{obs}}}$, independently of the record length.

Systems and protocol



Lorenz-63, $\lambda_{\max} t_{\mathrm{char}} = 0.68$



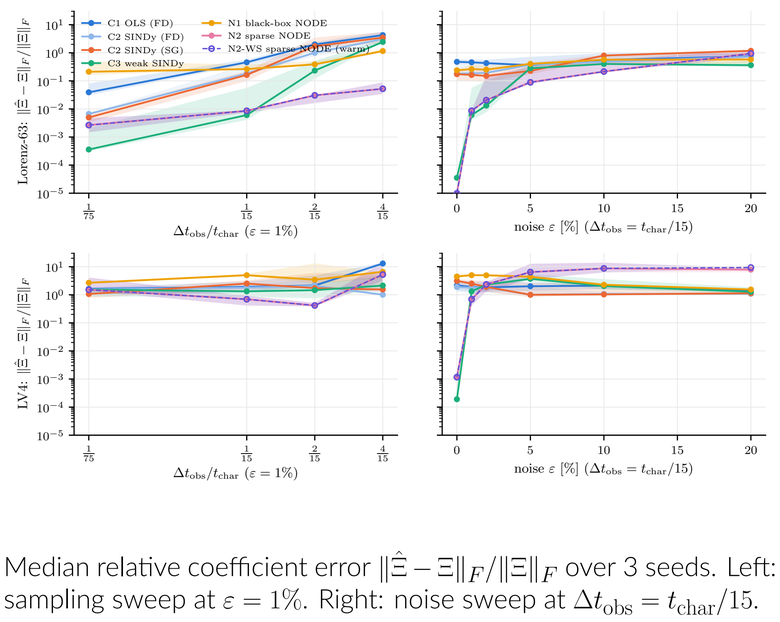
competitive LV ($d = 4$), $\lambda_{\max} t_{\mathrm{char}} = 0.67$

- 4 experiments, total $\frac{80}{3} t_{\mathrm{char}}$; $\Delta t_{\mathrm{obs}} / t_{\mathrm{char}} \in \{\frac{1}{75}, \frac{1}{15}, \frac{2}{15}, \frac{4}{15}\}$; $\varepsilon \in \{0, \ldots, 20\}\%$; 3 seeds.
- **C1** OLS **C2** SINDy (FD / Savitzky--Golay) **C3** weak SINDy
- **N1** black-box NODE + symbolic distillation **N2** sparse-library NODE
- All hyper-parameters are selected on held-out blocks; no estimator sees the true $\boldsymbol{\Xi}$.

N2: sparse-library neural ODE

1. Levenberg--Marquardt on the one-step flow map, $\Phi_{\Xi}^{\Delta t_{\mathrm{obs}}}(y_i) \approx y_{i+1}$.
2. L-BFGS on $\mathcal{J}_K$ over (Ξ, S) with continuation $K = 1, 2, 3, 5, 8, \ldots$
3. Multistart over $\{0,\ \Xi_{\mathrm{OLS}}\}$ (N2-WS: $+\Xi_{\mathrm{WSINDy}}$).
4. Sequential thresholding with trajectory-matching refits along a λ-path.

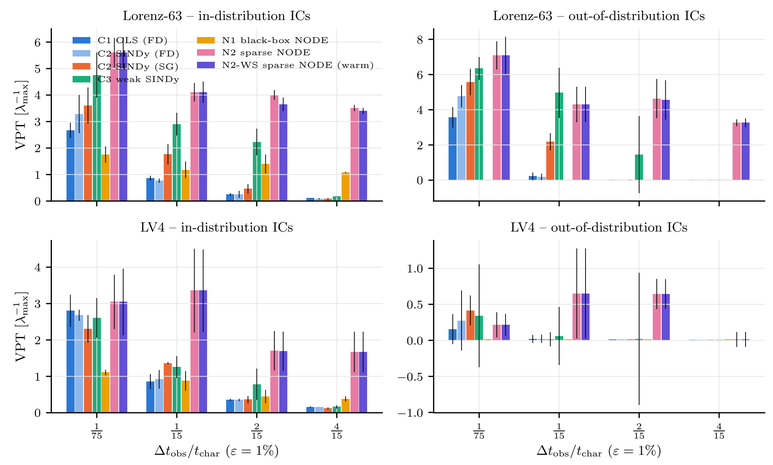
Benchmark: equation recovery



Median relative coefficient error $\|\Xi - \Xi\|_F / \|\Xi\|_F$ over 3 seeds. Left: sampling sweep at $\varepsilon = 1\%$. Right: noise sweep at $\Delta t_{\mathrm{obs}} = t_{\mathrm{char}}/15$.

- **Lorenz:** weak SINDy is best at dense sampling ($4 \cdot 10^{-4}$). The sparse NODE wins once $\Delta t_{\mathrm{obs}} \geq 2 t_{\mathrm{char}}/15$: 0.031 against 0.23, and 0.052 against 2.4.
- **LV4:** recovery only from noise-free data. The library's condition number on the attractor is $\approx 3 \cdot 10^4$, so 1% noise is already too much for *every* estimator.
- The distilled black-box equations are dense: error 0.21--1.15 on Lorenz.

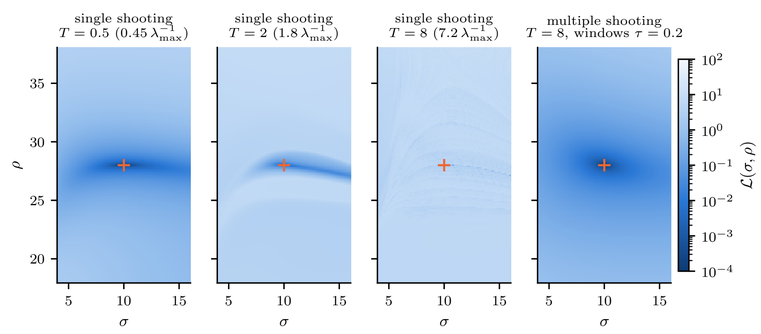
Forecasting: in- and out-of-distribution



Valid prediction time (normalised error < 0.4) in Lyapunov times, at $\varepsilon = 1\%$. The exact field reaches 11.3 (Lorenz) and 8.1 (LV4) with the same RK4 integrator.

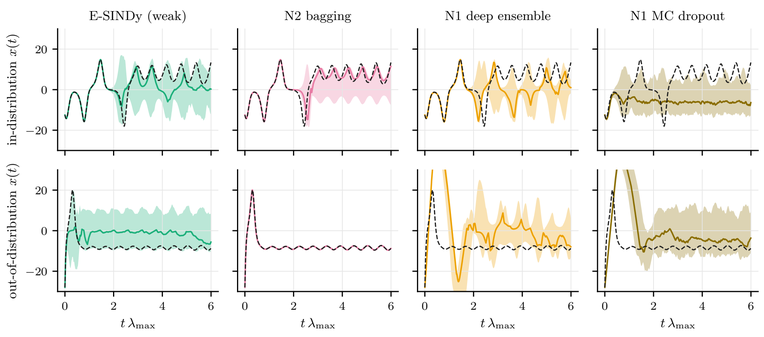
- The sparse NODE has the longest in-distribution VPT at every Δt_{obs}, on both systems: 3.5 at $\Delta t_{\mathrm{obs}} = 0.2$ on Lorenz, against ≤ 0.17 for the classical methods.
- **Out of distribution** (initial conditions $\geq 2\sigma$ from the attractor) the black-box NODE fails at once (VPT 0.015). Correct sparse models extrapolate (Lorenz N2: 3.3--7.1).

Loss landscapes and nonconvex optimisation



- Exact $\mathcal{L}(\sigma, \rho)$: single shooting becomes a flat surface with a needle-like minimum as T grows. Multiple shooting stays smooth.
- The median $\|\nabla \mathcal{L}_T\|$ grows like $e^{0.68T}$ ($\lambda_{\max} = 0.91$).
- 96 random restarts for (σ, ρ, β): single shooting with $T = 4$ succeeds 0% of the time, and multiple shooting with windows of 0.2 **100%**.
- Full library from random starts: continuation succeeds for 100% of them at $\Delta t_{\mathrm{obs}} = 0.05$ and for up to 100% at $\Delta t_{\mathrm{obs}} = 0.2$; direct training at the longest window succeeds for **0%**.

Uncertainty quantification



Ensemble forecasts, Lorenz with $\Delta t_{\mathrm{obs}} = 0.05$ and $\varepsilon = 1\%$. Top: in distribution. Bottom: out of distribution.

	coverage (90%) ID / OOD	CRPS ID / OOD	OOD/ID spread
E-SINDy (weak)	**0.87** / **0.93**	0.16 / 0.22	15.4
N2 bagging	0.18 / 0.11	**0.10** / **0.07**	3.9
N1 deep ensemble	0.41 / 0.47	0.51 / 1.35	11.9
N1 MC dropout	0.67 / 0.64	0.64 / 0.96	1.1

The bootstrap of the weak regression is calibrated, and window bagging is accurate but over-confident. Deep ensembles flag distribution shift; trajectory-wise MC dropout does not.

Take-aways

1. The gain of neural ODEs for discovery comes from **matching trajectories**, not from neural networks. The sparse-library NODE extends recovery to ~ 4 samples per period.
2. On chaotic flows that gain depends on **multiple shooting, continuation and multistart**. Single shooting fails beyond about one Lyapunov time.
3. Black-box NODEs give **no equations and no extrapolation**.
4. Identifiability is a property of the **data**: check $\sigma_{\min}(\Theta)$ before trusting any estimator.

References

Brunton, Proctor & Kutz, *PNAS* 2016 • Messenger & Bortz, *MMS* 2021 • Fasel et al., *Proc. R. Soc. A* 2022 • Chen et al., *NeurIPS* 2018 • Bock & Plitt, *IFAC* 1984 • Ribeiro et al., *Automatica* 2020 • Vano et al., *Nonlinearity* 2006 • Lakshminarayanan et al., *NeurIPS* 2017 • Gal & Ghahramani, *ICML* 2016 • Li et al., *NeurIPS* 2018